

Analytics & Insights — User Guide

How to run the project and read every chart on the Analytics page

September 2026

The **Analytics** page (`http://localhost:8000/analytics`) turns everything the WAF has recorded into charts, KPI cards and plain-language insights. Every number is calculated live from the project database `logs/detections.db` — nothing on the page is sample or made-up data.

This guide covers:

1. [Run the project](#) (end to end, from a fresh clone)
2. [Open the Analytics page](#)
3. [Add your own data](#) so the charts change
4. [Reading the page](#), section by section
5. [Filters, date ranges and comparisons](#)
6. [Exporting data](#) and the [analytics API](#)
7. [Troubleshooting](#)

Run the project

What you need

- **Python 3.11 or newer** (`python3 --version`)
- **Internet access** the first time (to install packages; PyTorch is about 200 MB)
- A modern browser (Chrome, Edge, Firefox or Safari)

Option A — one command (recommended)

From the repository root:

```
git clone https://github.com/kuldeep31016/Web-Application-Firewall.git
cd Web-Application-Firewall/transformer-waf

./scripts/setup_and_run.sh      # macOS / Linux
scripts\setup_and_run.bat      # Windows
```

The script does everything in order:

1. creates a virtual environment in `venv/`
2. installs `requirements.txt`
3. trains or downloads any model that is missing (first run only — the WAF model takes about 3 minutes, the URL classifiers about 20 minutes on a CPU)
4. starts the web service on **`http://localhost:8000`**

Leave the terminal open — the service runs until you press **Ctrl+C**.

Option B — step by step

```
cd Web-Application-Firewall/transformer-waf

# 1. virtual environment + packages
python3 -m venv venv
source venv/bin/activate          # Windows: venv\Scripts\activate
pip install -r requirements.txt

# 2. (optional) train the WAF model so new requests are scored and stored
PYTHONPATH=. python scripts/prepare_waf_data.py
PYTHONPATH=. python scripts/train_quick.py --epochs 40 --batch 64 --vocab 5000 \
    --embed 128 --heads 4 --layers 3 --ff 256 --maxlen 128 --lr 5e-4
PYTHONPATH=. python scripts/calibrate_threshold.py --benign-percentile 0.99 --write

# 3. start the service
PYTHONPATH=. python -m uvicorn src.api.detection_api:app --host 0.0.0.0 --port 8000
```

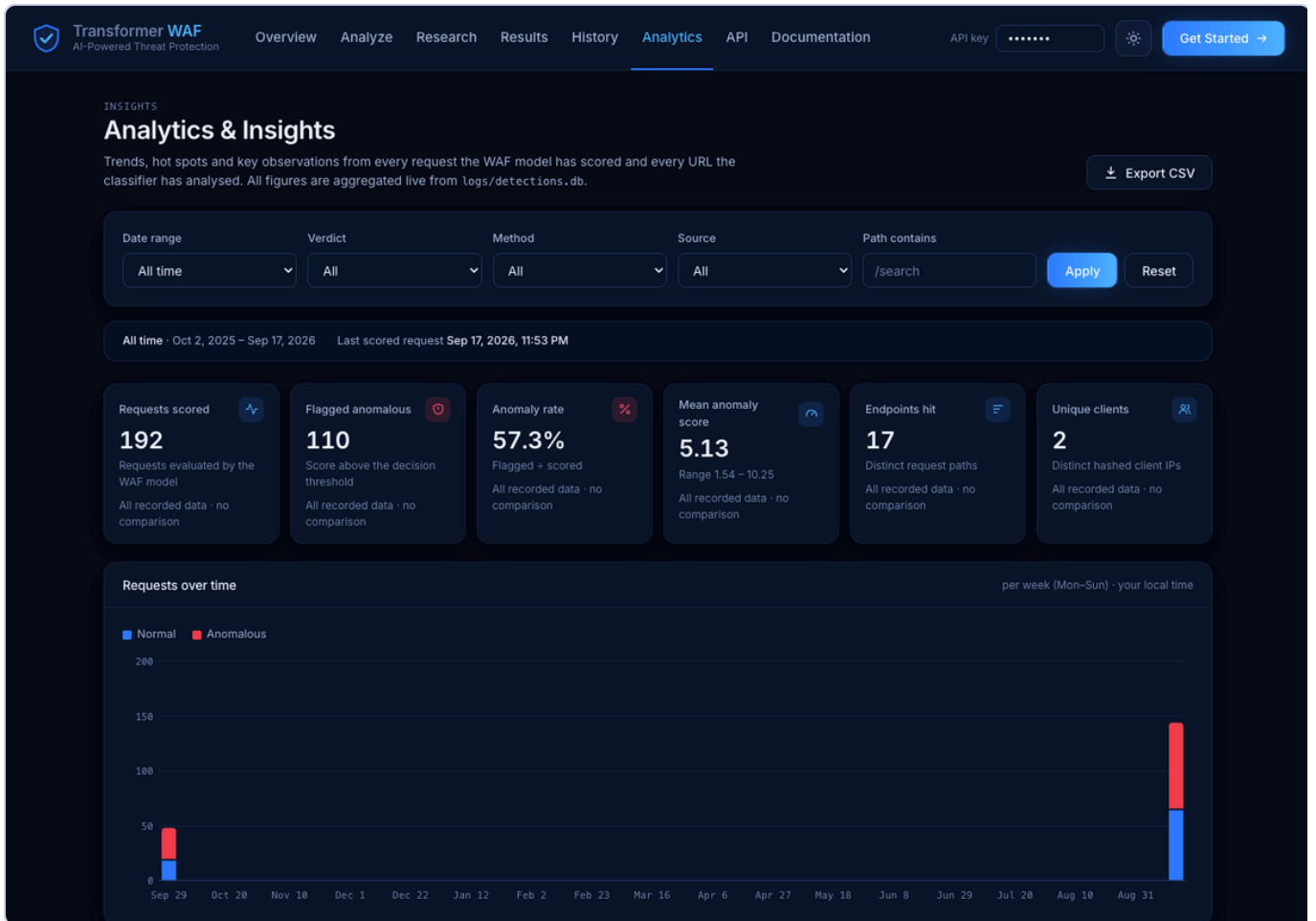
On Windows, set the path with `set PYTHONPATH=.` before the `python` commands.

Just want to look at the analytics? Step 2 is optional. The repository already ships with a `logs/detections.db` containing 192 scored requests and 22 URL analyses, so the Analytics page shows real data as soon as the service is running. Training is only needed to **add new** requests.

The service is ready when the terminal prints `Uvicorn running on http://0.0.0.0:8000`.

Open the Analytics page

1. Open **`http://localhost:8000/analytics`** in your browser (or click **Analytics** in the top navigation of any page).
2. The **API key** box in the top bar is pre-filled with `dev-key`, which is the default key. If the service was started with a different key (`WAF_API_KEY=...`), type that key in the box and press Enter.
3. The page loads with **Last 30 days** selected. Because the shipped data was recorded on specific dates, choose **All time** in *Date range* to see everything.



The top of the Analytics page: filters, range summary, KPI cards and the requests-over-time chart (dark theme, All time).

Add your own data

The analytics update every time the WAF scores something. After any of the actions below, change a filter or reload the Analytics page to see the new numbers. (These actions need the trained WAF model — see Option A or step 2 of Option B.)

Where	What to do	What appears in Analytics
Analyze page (<code>/analyze</code>)	Score an HTTP request or analyse a URL	A new request (source <i>Direct</i>) or a new URL analysis
History page (<code>/history</code>)	Click Replay on any stored request	A new request with source <i>Replay</i>
Command line	See the <code>curl</code> commands below	New requests (source <i>Direct</i> or <i>Batch</i>)

Score one request (a path-traversal attack):

```
curl -s -X POST http://localhost:8000/detect \  
  -H 'X-API-Key: dev-key' -H 'Content-Type: application/json' \  
  -d '{"method":"GET","path":"/download","query_params":{"file":"../../../../../etc/passwd"},"headers":{"body":""}}'
```

Score a normal request and an attack together (stored as source *Batch*):

```
curl -s -X POST http://localhost:8000/detect/batch \
-H 'X-API-Key: dev-key' -H 'Content-Type: application/json' \
-d ' [{"method":"GET","path":"/products","query_params":{"page":"1"},"headers":{},"body":""},
      {"method":"GET","path":"/search","query_params":{"q":"<script>alert(1)</script>"},"headers":{},"body":""}] '
```

Or run the ready-made attack suite (14 benign and malicious requests):

```
./scripts/run_tests.sh
```

Then open Analytics, pick **Today**, and the new requests appear in every chart.

Reading the page

Filter bar and range summary

The filter bar sits at the top; every chart, card and table below it follows these filters. The grey bar underneath states exactly which dates you are looking at, which period they are compared with, and when the last request was scored.

KPI cards

Six cards summarise the selected period:

Card	Meaning
Requests scored	How many HTTP requests the WAF model evaluated
Flagged anomalous	How many of them scored above the decision threshold
Anomaly rate	Flagged ÷ scored, as a percentage
Mean score	Average anomaly score, with the lowest–highest score underneath
Endpoints hit	Number of different request paths (e.g. <code>/search</code> , <code>/login</code>)
Unique clients	Number of different client IP addresses (stored only as a hash)

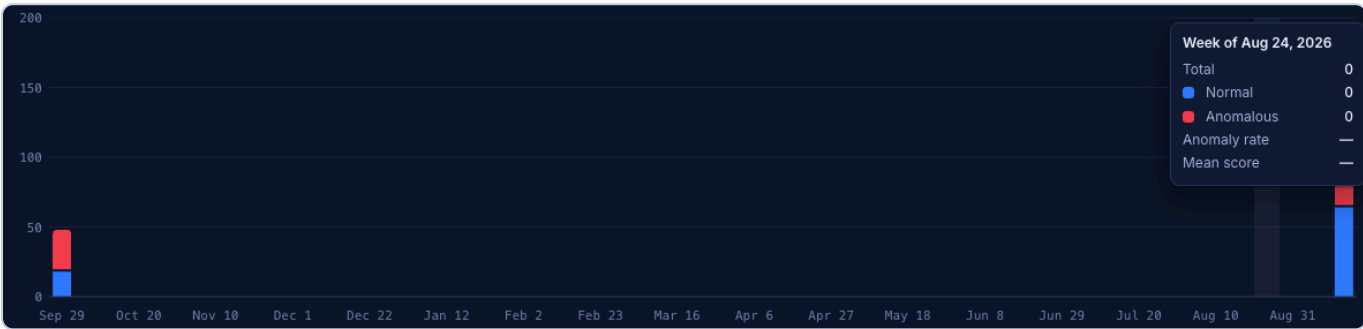
The last line of each card compares with the **previous period of the same length** (e.g. *Last 7 days* is compared with the 7 days before it):

- **↑ / ↓ with a percentage** — the change versus the previous period. The anomaly rate change is shown in **percentage points (pp)**.
- **Red** means a rise in something bad (more anomalies, a higher rate or score); **green** means it fell. Volume cards (requests, endpoints, clients) stay grey because more traffic is neither good nor bad by itself.
- **“New · none in previous period”** — the previous period had zero, so a percentage change is not meaningful.
- **“All recorded data · no comparison”** — shown when *All time* is selected.

Requests over time

A stacked bar chart: **blue = normal** requests, **red = anomalous** requests. Each bar is one hour, one day or one week — the page picks the size from the range length (up to 2 days → hourly, up to 120 days → daily, longer → weekly, Monday to Sunday). Bars use **your local time**. Days with no traffic are shown as empty slots, so gaps are real gaps.

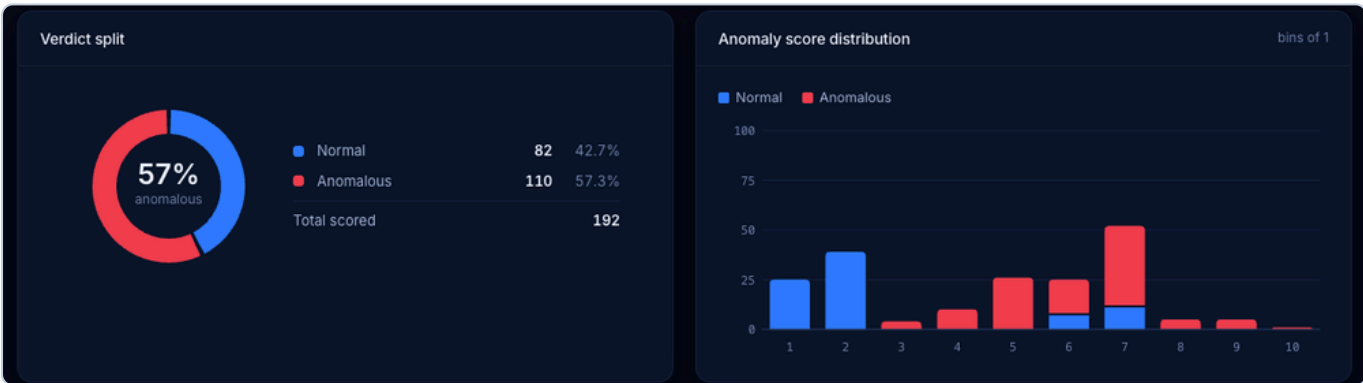
Hover over (or tap) any bar to see the exact figures for that period:



Tooltip on the requests-over-time chart.

Verdict split and score distribution

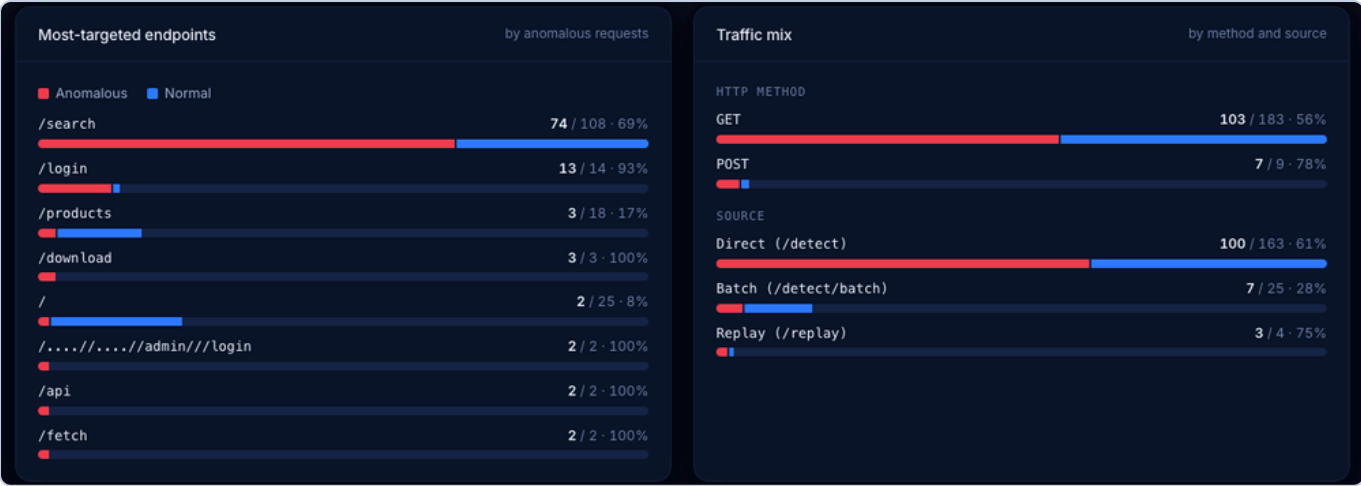
- **Verdict split** (donut): the share of requests that were normal vs anomalous, with counts and percentages beside it.
- **Anomaly score distribution** (histogram): how many requests fell into each score band. Normal requests should cluster on the left (low scores) and anomalies on the right. Where blue and red overlap, the model is less certain.



Verdict split and anomaly score distribution.

Most-targeted endpoints and traffic mix

- **Most-targeted endpoints** lists the paths that received the most anomalous requests. Each row shows *anomalous / total · anomaly rate*, and the bar is split red (anomalous) and blue (normal).
- **Traffic mix** shows the same split by **HTTP method** (GET, POST, ...) and by **source**: *Direct* (/detect), *Batch* (/detect/batch) or *Replay* (replayed from the History page).

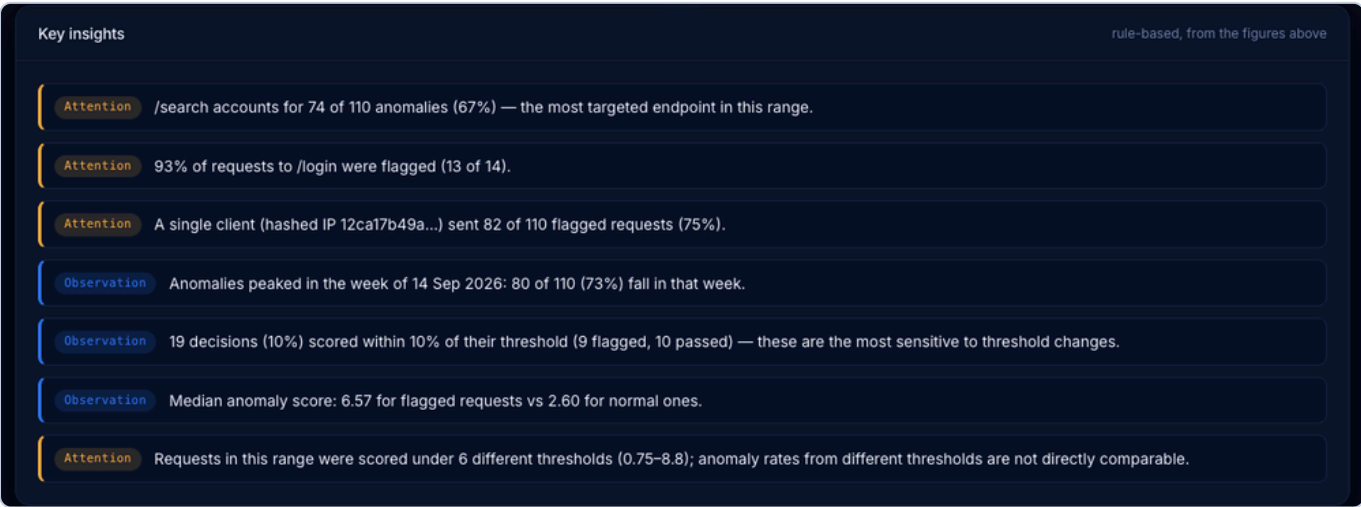


Most-targeted endpoints and traffic mix.

Key insights

Short sentences that summarise what matters in the selected data. They are produced by fixed rules from the numbers on the page (no AI guessing), and a rule only appears when the data supports it:

Label	Examples of when it appears
Attention (amber)	One endpoint receives $\geq 40\%$ of all anomalies; an endpoint has $\geq 50\%$ of its requests flagged; one client sends $\geq 50\%$ of flagged traffic; the anomaly rate rose by ≥ 5 pp; several thresholds were in use
Improving (green)	The anomaly rate fell by ≥ 5 pp
Observation (blue)	Traffic changed by $\geq 10\%$; the busiest hour/day/week for anomalies; how many decisions were close to the threshold; median score of flagged vs normal requests; endpoints attacked that were not attacked before; small-sample warning (< 20 requests)



Key insights for All time.

Score statistics and decision thresholds

- **Score statistics** gives minimum, median, mean, 95th percentile (P95) and maximum score for all requests, anomalous ones and normal ones. **Margin** is how far scores sit above (+) or below (–) their threshold on average. The note below counts decisions **within ±10 % of the threshold** — those would change verdict first if the threshold were adjusted.
- **Decision thresholds** lists each threshold that was active, how many requests it scored and when it was used. If more than one appears, anomaly rates from different periods are not strictly comparable.

Score statistics								Decision thresholds				
GROUP	N	MIN	MEDIAN	MEAN	P95	MAX	MARGIN	THRESHOLD	REQUESTS	FLAGGED	RATE	IN USE
All	192	1.54	5.44	5.13	8.12	10.25	+1.73	0.75	25	25	100.0%	Oct 2, 2025 – Sep 16, 2026
Anomalous	110	3.42	6.57	6.55	9.29	10.25	+3.69	8.8	6	0	0.0%	Oct 4, 2025
Normal	82	1.54	2.60	3.23	7.44	7.50	–0.90	8	3	0	0.0%	Oct 4, 2025
Near the threshold (within ±10%): 9 flagged and 10 passed — the decisions most sensitive to a threshold change. Margin is the mean distance above (+) or below (–) each request's own threshold.								7.5	6	0	0.0%	Oct 4, 2025
								7	12	9	75.0%	Oct 4, 2025
								3.0565	140	76	54.3%	Sep 16, 2026 – Sep 17, 2026
								Several thresholds were active in this range, so anomaly rates across time are not strictly like-for-like.				

Score statistics and decision thresholds.

Detailed breakdown table

A full table you can slice four ways using the tabs on the right: **Endpoint**, **Client**, **Method** or **Source**.

- **Sort** — click any column header; click again to reverse the order.
- **Search** — type in the box to filter rows (e.g. `api`).
- **Pages** — 10 rows per page; use **Previous / Next**.
- **Δ anomalous** — change in anomalous requests versus the previous period (hidden for *All time*).
- **Client** rows show the start of the hashed IP; *not recorded* means the request had no client address (e.g. replays).

Detailed breakdown

Endpoint

Client

Method

Source

Search

Filter rows...

17 rows

ENDPOINT	REQUESTS	ANOMALOUS ▾	RATE	MEAN SCORE	MAX SCORE	LAST SEEN
1. /search	108	74	68.5%	5.48	9.97	Sep 17, 2026, 11:53 PM
2. /login	14	13	92.9%	7.39	9.68	Sep 17, 2026, 01:07 AM
3. /products	18	3	16.7%	5.28	7.18	Sep 17, 2026, 01:07 AM
4. /download	3	3	100.0%	5.46	5.46	Sep 17, 2026, 01:07 AM
5. /	25	2	8.0%	1.97	8.41	Sep 17, 2026, 11:53 PM
6. /...../...../admin///login	2	2	100.0%	6.50	6.50	Sep 16, 2026, 10:21 PM
7. /api	2	2	100.0%	3.49	3.49	Sep 16, 2026, 10:21 PM
8. /fetch	2	2	100.0%	7.08	7.08	Sep 16, 2026, 10:21 PM
9. /ping	2	2	100.0%	8.12	8.12	Sep 16, 2026, 10:21 PM
10. /product	2	2	100.0%	3.42	3.42	Sep 16, 2026, 10:21 PM

Previous

page 1 of 2

Next

Detailed breakdown by endpoint.

URL classifier activity

Activity of the phishing URL classifier (every URL analysed on the Analyze page, or sent to `/detect/url` without `"persist": false`): how many URLs were analysed, how many were classified as phishing, average probability, inference time, how often parts of the URL were marked unavailable, and how often the WAF model also flagged the URL. The table breaks this down by handling strategy and model.

Only the **date range** applies here — method, verdict, path and source filters describe HTTP requests and do not apply to URLs.

URL classifier activity

date range applies; request filters do not

22

URLs analysed

all recorded data

63.6%

classified phishing

14 of 22

64.3%

mean P(phishing)

average model output

2.7 ms

mean inference time

per URL

16

with missing info

73% analysed with segments unavailable

4

also flagged by WAF

22 scored by the WAF model - 4 flagged by both

STRATEGY / MODEL	URLS	PHISHING	MEAN P	MS
none / baseline	15	11 (73%)	73.2%	3.2
augmented_training / robust	6	2 (33%)	36.2%	1.7
missing_token / baseline	1	1 (100%)	100.0%	1.0

INFORMATION UNITS MARKED UNAVAILABLE

scheme

15 analyses

subdomain

13 analyses

path

4 analyses

fragment

2 analyses

URL classifier activity.

Filters, date ranges and comparisons

Filter	Options
Date range	Today · Last 7 days · Last 30 days · Last 3 months · Last 6 months · This year · All time · Custom range
Verdict	All · Anomalous · Normal
Method	All · each HTTP method found in the data
Source	All · Direct · Batch · Replay
Path contains	Any text, e.g. <code>login</code> (press Apply or Enter)

- Drop-down filters apply immediately; **Reset** returns to *Last 30 days* with no filters.
- **Custom range** shows *From* and *To* date pickers. Both days are included in full (midnight to midnight, local time).
- The **previous period** is always the same length immediately before the selected one. *All time* has no previous period.
- The page remembers your last date-range choice in this browser.

Exporting data

Click **Export CSV** (top right). The file contains the **Detailed breakdown** exactly as currently filtered, grouped (Endpoint / Client / Method / Source), searched and sorted — every row, not just the visible page. It includes requests, anomalies, anomaly rate, mean and max score, last seen (UTC) and, when a previous period exists, the previous-period figures.

Analytics API

The page is powered by three endpoints that you can also call directly. All require the `X-API-Key` header, like every other endpoint.

Endpoint	Returns
GET <code>/analytics/overview</code>	KPIs, comparison, time series, distributions, score statistics, top endpoints, URL activity and insights
GET <code>/analytics/breakdown</code>	One page of the breakdown table (<code>group_by</code> , <code>sort</code> , <code>order</code> , <code>search</code> , <code>limit</code> , <code>offset</code>)
GET <code>/analytics/export</code>	The whole breakdown as a CSV file

Common parameters: `start` and `end` (Unix seconds; omit both for all time), `tz_offset` (your UTC offset in minutes, e.g. `330` for India), `verdict` (`anomaly` / `normal`), `method` , `source` (`direct` / `batch` / `replay`), `path` .

```
# all-time overview
curl -s -H 'X-API-Key: dev-key' http://localhost:8000/analytics/overview

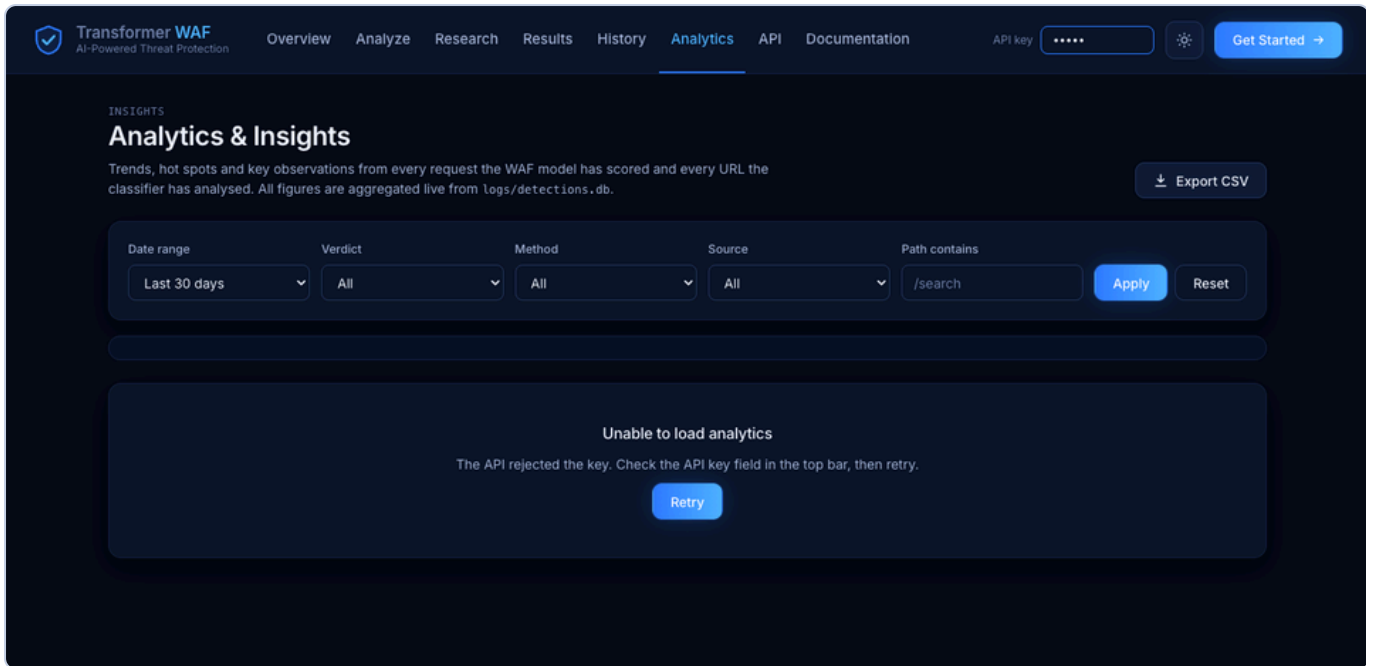
# top 5 endpoints by anomalies
curl -s -H 'X-API-Key: dev-key' \
  'http://localhost:8000/analytics/breakdown?group_by=path&sort=anomalies&limit=5'

# breakdown by client as CSV
curl -s -H 'X-API-Key: dev-key' -o clients.csv \
  'http://localhost:8000/analytics/export?group_by=client'
```

Interactive documentation for all endpoints is at **<http://localhost:8000/docs>** (click *Authorize* and enter `dev-key`).

Troubleshooting

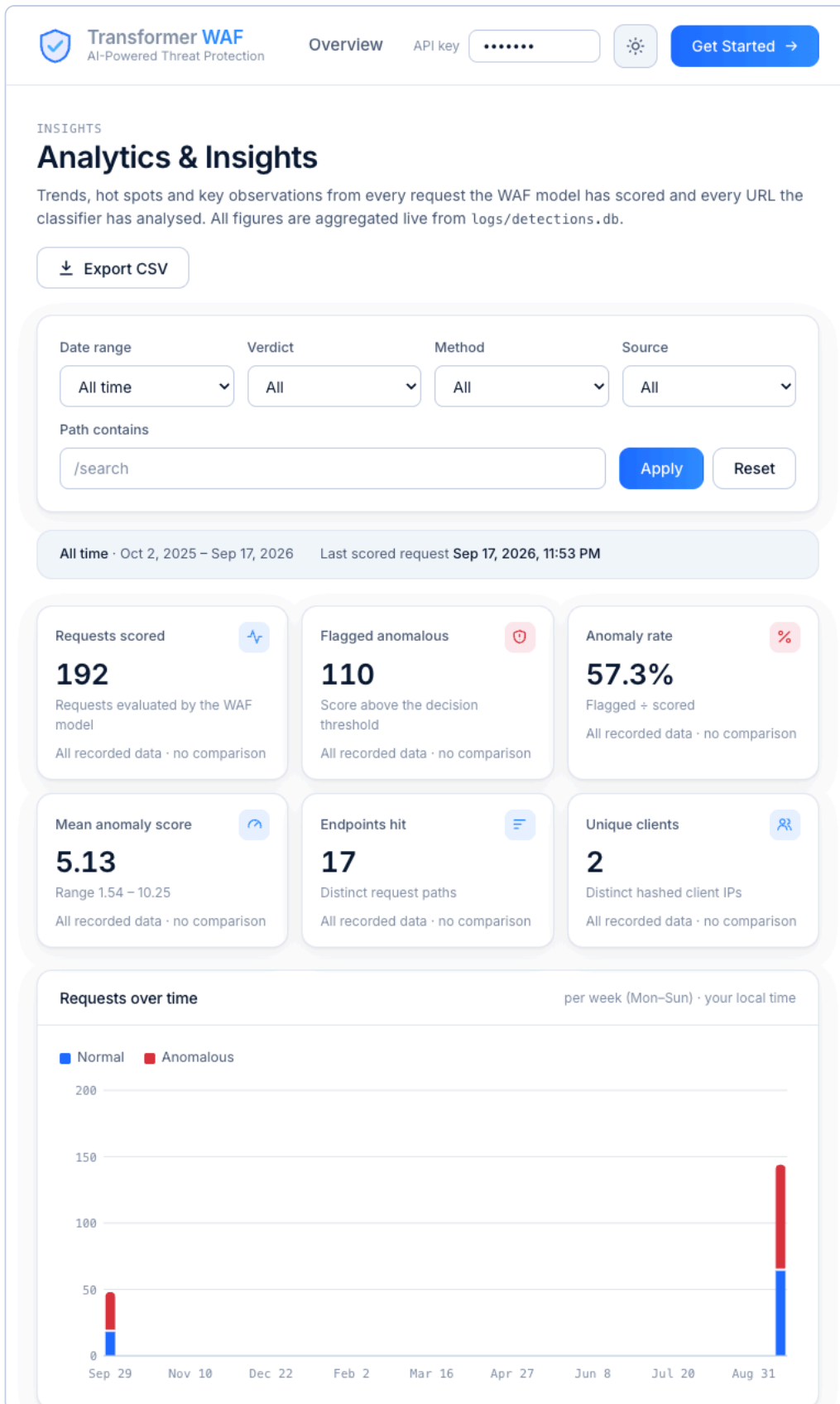
Problem	Fix
Browser says <i>can't connect</i>	The service is not running. Start it (Option A or B) and keep the terminal open.
“Unable to load analytics — The API rejected the key”	Type the correct key (default <code>dev-key</code>) in the <i>API key</i> box in the top bar, press Enter, then Retry .
“No analytics data available”	Nothing matches the date range and filters. Choose All time or press Reset .
New requests do not appear	The WAF model is not trained, so <code>/detect</code> returns placeholder answers without storing them. Run <code>./scripts/setup_and_run.sh</code> (or step 2 of Option B), restart, and try again.
URL analyses do not appear	The URL classifiers are not trained yet (the script trains them), or the URL was sent to <code>/detect/url</code> with <code>"persist": false</code> (the Overview page demo does this on purpose).
Address already in use	Port 8000 is taken. Stop the other program or start with <code>PORT=8010 ./scripts/setup_and_run.sh</code> , then open <code>http://localhost:8010/analytics</code> .
Charts look cramped	Rotate the phone or widen the window — charts redraw to fit the screen.



The error state shown when the API key is wrong; Retry reloads after you fix the key.

Themes and devices

The page follows the site theme: use the sun icon in the top bar to switch between dark and light. It works on desktop, tablet and phone — cards re-arrange into two columns and charts resize to the screen width.



Transformer WAF

AI-Powered Threat Protection

.....

Get Started →

Overview

Analyze

Research

Results

INSIGHTS

Analytics & Insights

Trends, hot spots and key observations from every request the WAF model has scored and every URL the classifier has analysed. All figures are aggregated live from logs/detections.db.

↓ Export CSV

Date range

Verdict

All time

All

Method

Source

All

All

Path contains

/search

Apply

Reset

All time · Oct 2, 2025 – Sep 17, 2026

Last scored request Sep 17, 2026, 11:53 PM

Requests scored

192

Requests evaluated by the WAF model

All recorded data · no comparison

Flagged anomalous

110

Score above the decision threshold

All recorded data · no comparison

Anomaly rate

57.3%

Flagged ÷ scored

All recorded data · no comparison

Mean anomaly score

5.13

Range 1.54 – 10.25

All recorded data · no comparison

Endpoints hit

17

Distinct request paths

All recorded data · no comparison

Unique clients

2

Distinct hashed client IPs

All recorded data · no comparison

Requests over time

per week (Mon–Sun) · your local time

■ Normal

■ Anomalous

200

150

Phone layout (375 px).

Good to know

- **No accuracy figures for live traffic.** Real requests have no “correct answer” attached, so the page shows what the model decided, not whether it was right. Accuracy, precision and recall for the models are on the **Research** page, measured on labelled test data.
- **Daylight saving.** Hour and day bars use your current UTC offset; in a range that crosses a daylight-saving change, bar edges can shift by one hour.
- **Client privacy.** Client IP addresses are never stored in plain text — only a hash — and the page shows only the first characters of it.